```
# Install spaCy
!pip install spacy

# Download English model
!python -m spacy download en_core_web_sm

# Import spaCy
import spacy

# Load the model
nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")
```

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ied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
ied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
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ied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
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ied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
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ied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
ied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
ied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
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ied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
ied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
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ied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8
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ge via spacy.load('en_core_web_sm')
dencies
 Colab notebook, you may need to restart Python in
```

```
age's dependencies. You can do this by selecting the
rt runtime' option.
fully
```

```
sentences = [
"The student completed the assignment.",
"The teacher explained the lesson clearly.",
"She bought a new laptop.",
"The boy kicked the ball.",
"The dog barked loudly.",
"The manager approved the project and sent an email.",
"I finished my homework before dinner.",
"Although it was raining, the match continued.",
"The professor who teaches AI is very kind.",
"She studies hard because she wants good grades.",
"Did you complete the report?",
"Where is the nearest hospital?",
"The project was completed by the team.",
"The cake was baked by my mother.",
"The email was sent yesterday."
]
```

```
docs = [nlp(sentence) for sentence in sentences]
```

```
import pandas as pd

data = []

for doc in docs:
    for token in doc:
        data.append([
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])

df = pd.DataFrame(data, columns=["Token","Lemma","POS","Dependency","Head"])
print(df)
```

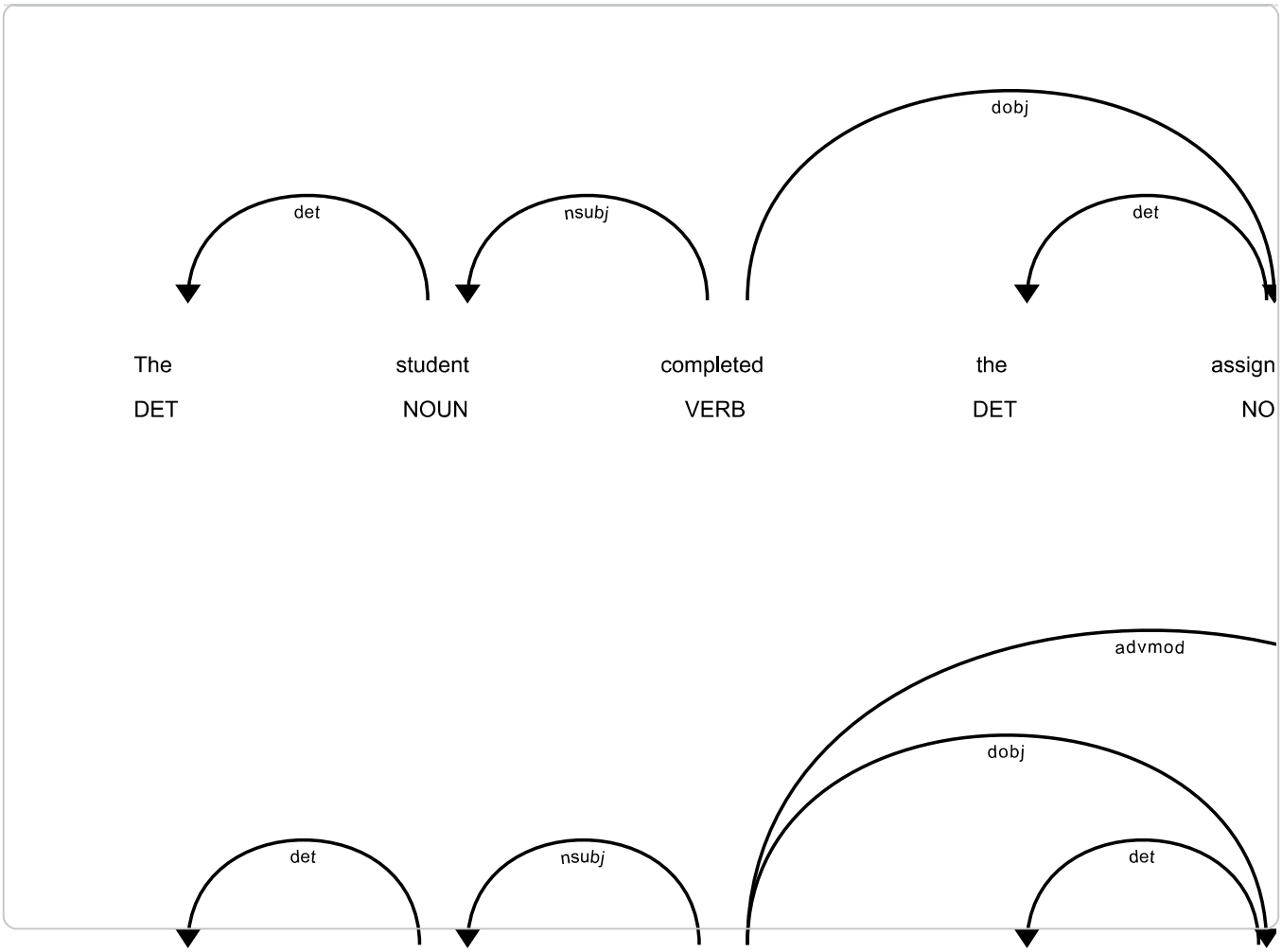
```
          Token       Lemma    POS Dependency        Head
0           The         the    DET        det     student
1       student     student   NOUN      nsubj   completed
2     completed    complete   VERB       ROOT   completed
3           the         the    DET        det  assignment
4    assignment  assignment   NOUN       dobj   completed
..          ...         ...    ...        ...         ...
103       email       email   NOUN  nsubjpass        sent
104         was          be    AUX    auxpass        sent
105        sent        send   VERB       ROOT        sent
106   yesterday   yesterday   NOUN   npadvmod        sent
107           .           .  PUNCT      punct        sent

[108 rows x 5 columns]
```

```
from spacy import displacy

for sentence in sentences[:5]:
    doc = nlp(sentence)
```

```
displacy.render(doc, style="dep", jupyter=True)
```


dobj
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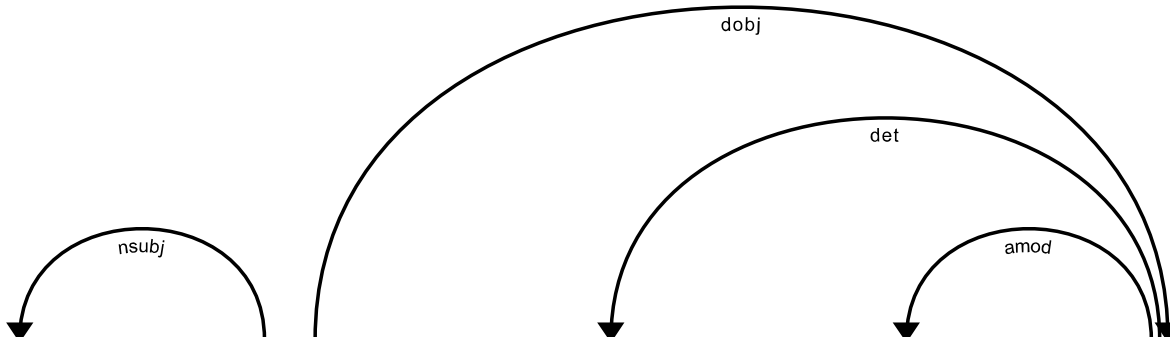
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